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Optimization of inter-basin water transfer projects: from macroeconomic impacts to technical hydraulic design

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Water scarcity and supply insecurity are the primary barriers to economic growth in arid and semi-arid regions. Inter-basin water transfer projects are high-cost solutions that could create socio-environmental side effects. This study presents a novel approach to optimally design such projects from an investor's perspective. We employed an energy input–output analysis-based linear programming (LP) model to determine the water demands of various economic sectors within an optimal development scenario. The results from this LP model, combined with a survey on the sectors' willingness to pay, are then integrated into a mixed-integer non-linear programming (MINLP) model formulated to optimally design the long-distance Khersan Dam-to-Kerman Province water conveyance project. The project is a 650-km long water transfer line with a cumulative pumping height of around 1500 m whose cost is sensitive to its capacity to be optimized. The model optimizes the project's capacity based on the level of investment economically justified by the sectors demanding a sustainable water supply. Our findings indicate that approximately 85 percent of the increase in the water demand of Kerman's economic activities (except agriculture, water provision, and service sectors) over the project's lifetime could be met using only 5.2% of its total capacity. Furthermore, the financial participation of these economic activities could enhance the project's net benefit by up to 6.3%.

Keywords Optimization, Water conveyance, Input–output, Cost–benefit analysis

The uneven spatial and temporal distribution of Earth's freshwater resources presents a fundamental challenge for water resources management. According to the United Nations, approximately 343 million people worldwide (5.8 percent of the global population) living in arid and semi-arid regions face water supply insecurity due to climate change and global warming[1]. Water resources management under drought conditions necessitates a transition from standard operating procedures to adaptive strategies that balance anthropogenic demand with ecological preservation. Policies to improve water security generally fall into two categories: a) supply-side policies, such as dam construction to address temporal unevenness and water transfer projects to address spatial unevenness, and b) demand-side policies, such as water markets, economic incentives (e.g., subsidies, taxes, and tariffs), and consumer education. The literature consistently prioritizes demand management over supply augmentation[2–5]. Nevertheless, supply-side solutions can become inevitable in cases of severe water scarcity.

Inter-basin water transfer is a supply-side augmentation option designed to alleviate the uneven spatial distribution of water resources[6,7]. Globally prominent examples of such infrastructural megaprojects include China's South-to-North Water Diversion[8], the California State Water Project in the United States[9], and the transboundary Lesotho Highlands Water Project in Southern Africa[10]. In recent years, increasing attention has been devoted to integrated water transport and abstraction management strategies aimed at safeguarding critical water reservoirs that provide potable water and essential ecological services. In some cases where the water crisis is less severe, solutions such as allocation management can provide adaptation to water scarcity conditions. E.g, Lake Mead utilizes a tiered system where water delivery reductions are triggered by specific elevation thresholds, to manage systemic shortages in the Colorado River Basin[11,12]. Conversely, sever water shortage crises might require a combination of abstraction controls and large-scale water conveyance projects (WCPs). A prominent example is Lake Urmia, which lost over 80% of its surface area due to severe droughts and unsustainable

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anthropogenic withdrawals, resulting in hazardous salt storms. To combat this ecological disaster, the Zab River water transfer project was implemented. This megaproject systematically diverts approximately 600 million cubic meters of water annually to guarantee environmental flows, and stabilize the lake's water levels[13].

The high costs and significant socio-environmental consequences of WCPs typically relegate these initiatives to one of the last resorts in water resource management hierarchies. Some scholars further contend that the legitimacy of such projects lies primarily in their capacity to supply drinking water, while other uses are viewed as less justifiable[14]. The case of Kerman Province in Iran is one such example. Kerman is the largest province in Iran, with a total area of 183,285 sq. km, and is located in an arid region with a mean annual precipitation of 134 mm[15]. Groundwater resources provide 87 percent of the potable water and 95 percent of the total water supply in Kerman[16]. However, the province's main aquifers have experienced drastic declines due to over-exploitation, with an average water level drop of more than 0.5 m per year over the past two decades. Consequently, water supply security for all purposes, especially for drinking water which requires higher quality and reliability, has been seriously endangered[2].

The water crisis in Kerman has prompted government action, leading to plans for the construction of a costly WCP known as the Khersan Dam Conveyance Line (KDCL). The KDCL is designed to transfer water from the Karun Basin to Kerman over a distance of 640 km. Although there is a consensus that the KDCL is a practical solution to Kerman's water shortage, stakeholders have yet to agree on the volume of water to be transferred. A key dispute revolves around the necessity of supplying water for productive sectors in addition to meeting drinking and sanitation needs. Policy-makers are seeking to share the cost of KDCL with the productive sectors, but the financing of the KDCL present a dilemma. On one hand, the project involves high fixed costs, which implies a declining unit cost of water as the project's scale increases. On the other hand, the enormous expense of water supply via the KDCL could render the expansion of water-intensive industries either practically infeasible or economically unjustified. Therefore, determining the optimal scale of the project and the corresponding volume of water to be transferred remains a critical question.

Several studies have examined inter-basin WCPs from technical[8], social[17], environmental[18], legal[19], and economic[20] perspectives. Han, et al.[8] developed a Copula-Bayesian coupling framework to examine how often and under what circumstances wet versus dry conditions in the source basin encounter matching or mismatching hydrological states in the receiving basin. Wei, et al.[17] applied game theory to simulate and resolve water allocation and nitrogen reduction conflicts in the Middle Route of China's South-to-North water transfer project. Ding, et al.[18] quantified the impact of the Yangtze-to-Huaibe Water Diversion Project's construction on land use and the supply–demand balance of Ecosystem Services in the Anhui section. Wang and Li[19] Examined how China's institutions have adapted (or failed to adapt) to manage the middle route of the South-to-North Transfer, with emphasis on legal frameworks, governance, institutional fragmentation, and policy mandates. Cao, et al.[20] developed a multi-scenario optimization model to improve the operation of inter-basin water transfer systems by explicitly balancing transfer costs and supply reliability.

Among the studies evaluating WCPs from an economic perspective, many emploied simulation–optimization approaches to perform cost–benefit analyses (CBA)[5,21–24]. Draper, et al.[25] proposed an economic–engineering optimization framework for the California water supply system, demonstrating the practicality of optimization models with economic objectives for large-scale water resources management. To achieve a more comprehensive CBA, researchers have attempted to quantify the costs of environmental and social impacts of WCPs by incorporating them into the analysis as additional economic variables[6,20,24]. However, the economic benefits of WCPs are still often restricted to revenues from water sales to specific, predefined users—typically farmers[6,23,24,26]. E.g, Nasiri-Gheidari, et al.[24] developed a multi-objective optimization model to identify an optimal water transfer pattern that minimizes environmental pollution while maximizing economic benefits for the initial plan of the KDCL project; yet their analysis assumes agriculture as the sole beneficiary of the transferred water.

While a number of studies have applied economic models—such as computable general equilibrium (CGE) and energy input–output analysis (EIOA) frameworks—to assess the economic impacts of WCPs on donor and recipient regions[21,22,27–29], they rarely address the optimal hydraulic design of the conveyance system[30,31]. simultaneous consideration of both aspects is essential for developing an optimal WCP plan, as the unit cost of water transfer depends directly on the project's scale and capacity, whereas the transfer volume should be determined by demand-side factors, including customers' willingness to pay (WTP)[32]. Accordingly, outputs from the economic model should inform a conveyance-line optimization model to determine the project's optimal capacity.

This study employs an EIOA model to incorporate the economic structure of the receiving region of the WCP and resource consumption scenarios[33] into the optimization of the KDCL project design. Using an EIOA-based linear programming (LP) model, water demand for different economic sectors is determined under the optimal development scenario, i.e., maximizing economic growth in Kerman province. The EIOA model also estimates the water intensity of each economic sector, which is combined with survey data from production units to calculate their willingness to pay (WTP) for water. These water demand and WTP values are used to construct a discrete benefit function for different volumes of transferred water. Recognizing that the volume of water transfer influences both the conveyance line design and the unit cost of water provision, this study considered mutual effect of them to provide a more comprehensive solution. Therefore, the resulting benefit function is incorporated into a MINLP model to determine the cost of water provision and identify the optimal design of the WCP.

The paper is structured as follows. The following section of Methods and Materials introduces the case study, and shows how data required for developing the EIOA model are collected and used. It then explains how the EIOA and cost estimation models are consolidated within an optimization problem framework; then it presents the formulation of the optimization problem developed based on the cost estimation equations and

the outcome of the EIOA model. The results and discussion section provides the outputs of the optimization problem, including the optimal design of the project and optimal water transfer plan. Finally, the conclusion section summarizes all discussions and provides suggestions for future studies.

Methods and materials

Case study

With a total area of 183.2 km^2, Kerman Province is the largest province of Iran. Similar to other provinces located in the eastern strip of the country, it suffers from severe water shortages. According to the 2016 population census, Kerman's population is 3.2 million and contributes 3.7% to Iran's GDP (excluding oil), with the industry, agriculture, and mining sectors producing 1.76, 1.9, and 0.7 billion USD, respectively[34]. Water scarcity is one of the main barriers to economic development in Kerman[35]. Kerman's water supply relies on local surface and groundwater resources, providing 10.5% and 89.5% of the total water supply, respectively, i.e., 5598 MCM/year[16]. The groundwater level in Kerman has declined by approximately 0.5 m per year over the past two decades. Similarly, surface runoff has decreased by more than 35% compared to its long-term average (IWPC, 2020). In addition, groundwater quality has been poor and continues to deteriorate; the concentration of heavy metals—particularly arsenic—in some aquifer sections is ten times higher than the drinking water standard (INWW Eng. Co., 2020). These conditions indicate that Kerman's local water resources will be insufficient to meet even the drinking water demand, estimated to increase by 1.6% annually and reach 561 MCM by 2045, neither quantitatively nor qualitatively.

Agricultural activities consume about 92% of the total water supply, and agricultural water use has remained almost constant over the past decade. Beyond water quality concerns, meeting drinking water demand by reducing agricultural withdrawals is highly challenging, largely due to the difficulty of persuading local farmers. Consequently, the KDCL project has become a government priority in the region. As depicted in the spatial layout of the KDCL project (Fig. 1), the 650-km conveyance line traverses substantial topographical obstacles, extending from the Karun Basin through the Zagros Mountain Range to Kerman (Fig. 4). Notwithstanding an elevation differential of approximately 100 m between the origin and destination sites, a cumulative pumping head of roughly 1,500 m is required to facilitate water transfer along the conveyance path. This configuration necessitates a hydraulic design incorporating multiple high-head pumping stations and pressure-reducing valves (PRVs). Furthermore, the deployment of pipelines with higher pressure classes is imperative. While the primary purpose of the KDCL is to supply drinking water, it could secure water supply for the economic activities, except agriculture and services, with an annual water need of only around 55 MCM and a high share of gross domestic production (GDP) (63 percent of Kerman's GDP)[34]. Thus, the project has the potential to act as a driver of regional economic development. Moreover, supplying water to economic activities could partially finance this high-cost project.

Modeling approach

The methodology adopted in this study involves quantifying the costs and benefits of water supply to different economic sectors and then solving a combinatorial nonlinear optimization model to determine both the optimal volume of water transfer and the project's design variables, including pipeline dimensions and length, the

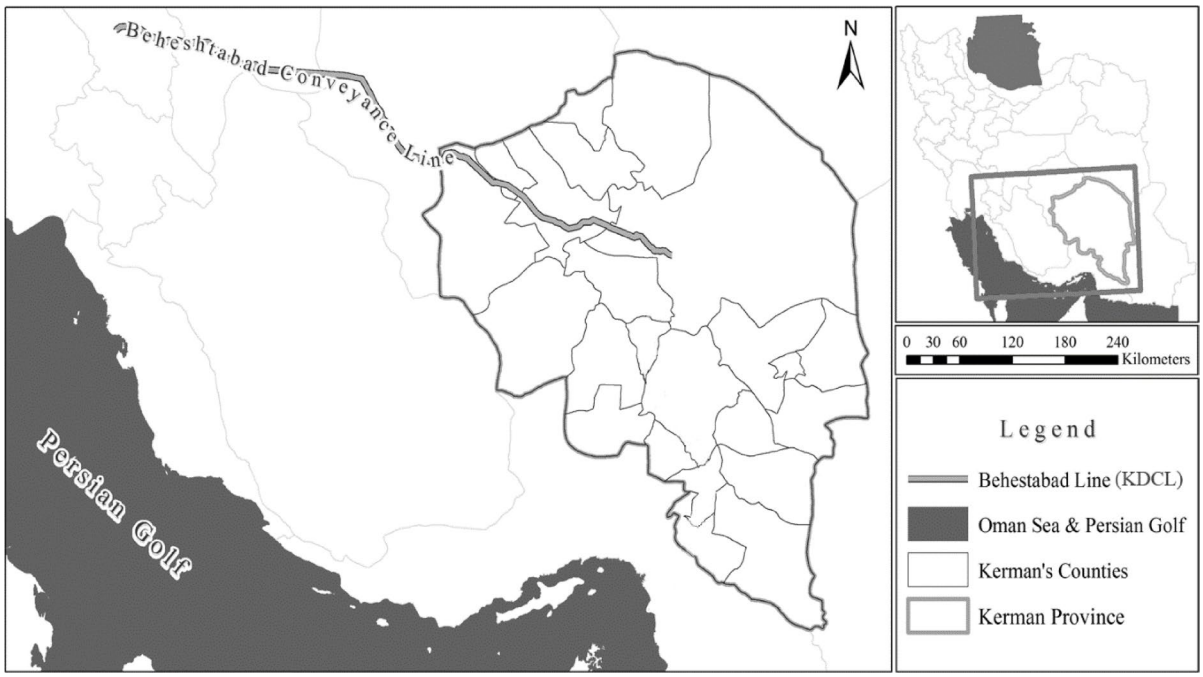


Fig. 1. Map of the initial conveyance line proposed by IWPC.

capacity and location of pumping stations, the placement of pressure reduction valves (PRVs), pipe's working pressure and flow velocity. The model's objective function maximizes the net benefit of the project, defined as the difference between the water price in the donor and receiving regions minus the cost of water provision. While the unit cost of water provision generally decreases with project scale, the WTP of water users declines as the transfer volume increases[36]. Clearly, the optimal solution is achieved when the marginal cost of water provision equals the marginal WTP of consumers.

Figure 2 illustrates the procedure used to derive the optimal plan for the KDCL project. The proposed model comprises two separate modules, i.e., cost and benefit estimation modules, which are incorporated simultaneously into the optimization model. In the cost estimation module, several variables and parameters, including project scale and electricity price, influence the design of the conveyance line and, consequently, the unit cost of water provision. These variables and parameters are functionally dependent and interact through governing flow equations and design criteria. To capture these relationships, a simulation model was developed based on the KDCL project's initial plan proposed by Iran Water and Power Resources Development Co. (IWPC), as the KDCL project client. Using defined technical standards, construction price lists, and design equations, simulation model estimate the cost of water provision at different transfer volumes.

The benefit estimation module quantifies the economic gains from water sales by incorporating both the economic sectors' WTP for water and their projected water demand. The projected water demand of each sector was estimated in accordance with their expected expansion over the project's lifetime. An EIOA-based linear programming model (Eqs. 7 to 11) was employed to identify the optimal development pattern under water scarcity, determining the expansion trajectory of the economic sectors during the project operation period. The WTP of the sectors was estimated using a function fitted to data from selected production units, linking their water use intensity with WTP. These data were obtained from contract records provided by the Iran National Water and Wastewater Engineering Company, where the prices agreed upon in water sales contracts served as the basis for calculating WTP. The outcomes of these estimations were used to derive the relationship between water supply and the corresponding benefit was derived and incorporated as a discrete function into the optimization model presented in the following section. It is important to note that the proposed methodology is developed based on the following key assumptions. First, the significant decline in groundwater levels is assumed to constrain further abstraction from local sources, implying that any additional water demand must be satisfied through the KDCL project. Second, a portion of the drinking water supply is anticipated to be provided via seawater desalination.

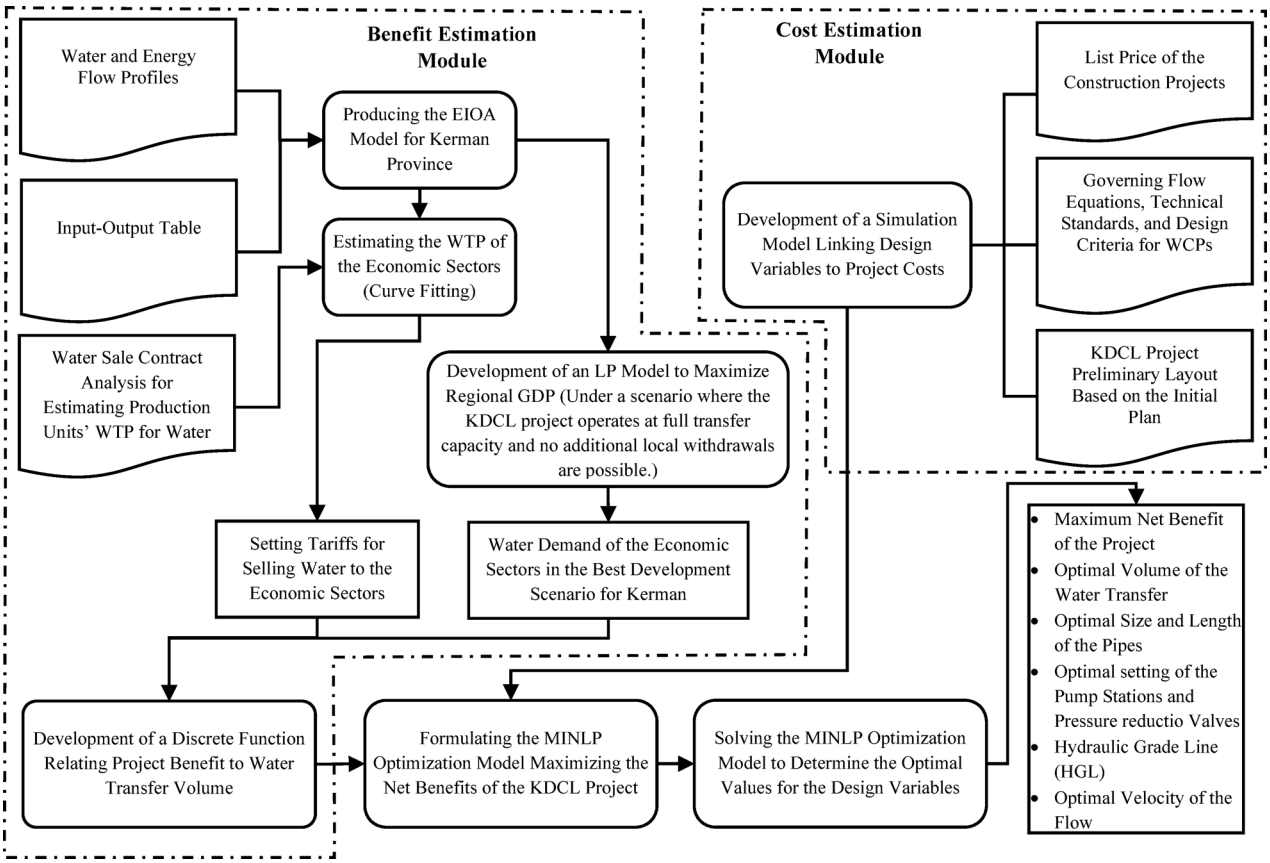


Fig. 2. Flowchart of the methodology for determining the optimal water transfer in the KDCL project.

Optimization model

The optimization model proposed in this study involves a simple CBA (Eqs. 1 to 4) to assess the cost-effectiveness of supplying water to Kerman's economic activities. The term $Benefit_{Total}$ in the model is calculated using the benefit estimation module (Eqs. 5 to 8), which incorperate the demands estimated by the EIOA-based LP model and water sale tariffs determined according to WTP of the economic sectors to estimate revenues for different water transfer volumes. Water transfer volume affects the project costs, including the implementation ($Cost^{Imp}$) and operation ($Cost^{Opr}_{PV}$) costs, calculated via the cost estimation module (Eqs. 16 to 35). By simultaneously solving the cost and benefit equations, the model determines the optimal water transfer volume and other design variables. The presence of discrete variables and non-linear relationships in this model converts it to a MINLP, an NP-hard combinatorial optimization problem.

$$Max \ \ \mathrm{NB_{KDCL}} = Benefit_{Total} - Cost_{Total} \tag{1}$$

Subject to:

$$Cost_{Total} = Cost^{Imp} + Cost^{Opr}_{PV} \tag{2}$$

$$Cost^{Opr}_{PV} = \sum_{\boldsymbol{t}=1}^{25} \frac{Cost^{Opr}(\boldsymbol{t})}{(1+\mathbf{IR})^{\boldsymbol{t}}} \tag{3}$$

$$Benefit_{Total} = \sum_{\boldsymbol{t}=1}^{25} \frac{Benefit(t)}{(1+\mathrm{IR})^{t}} \tag{4}$$

where $\mathrm{NB_{KDCL}}$ is the net benefit of the KDCL project, $Cost_{Total}$ is the project's cost, including the implementation cost ($Cost^{Imp}$), obtained by solving Eqs. 16–21, plus the present value of the operation cost ($Cost^{Opr}_{PV}$), and $Benefit_{Total}$ is the present value of the benefits over the project's lifetime.The index t is the number of years elapsed from the beginning of the project, and $Cost^{Opr}(t)$ and $Benefit(t)$, determined by solving Eqs. 22–35 and Eqs. 5–8 respectively, denote the operation costs and the revenues from water sales in year t. IR (equivalent to 8%) is the capital return rate used in the present value calculations.

Benefit estimation module

Equations 6 to 8 form the benefit estimation module of the optimization model developed to maximize the net benefit of the KDCL project. As outlined earlier, project benefits include revenues from water sales to economic sectors, priced according to their marginal willingness to pay (WTP), and revenues from potable water supplied to households at the regulated tariff (Eq. 5). Equation (6) ensures that the quantity of water sold to each economic sector does not exceed its estimated demand. The procedure for estimating sectoral water demand ($\mathrm{Demand}(k,t)$) and WTP is elaborated in detail later in this section. To prevent production capacity expansions planned in the previous year or earlier from remaining unused, Eq. 7 was incorporated into the model to ensure that water allocations are at least equal to those of previous years. Finally, the total water transferred by the KDCL equals the sum of water sold to economic sectors and that allocated for drinking purposes, as defined in Eq. 8. The mathematical program is therefore formulated as follows:

$$Benefit(t) = 365 \times 86400 \times (\mathrm{Q_{min}} \times \mathrm{p_{pot}} + \sum_{k=1}^{14} Q(k,t) \times \mathrm{p}(k)) \quad \forall \boldsymbol{t} \in \mathbf{1},\ldots,\mathbf{25} \tag{5}$$

$$Q(k,t) \le \frac{\mathrm{Demand}(k,t)}{365 \times 24 \times 3600} \quad \forall k \in 1,\ldots,14 \quad \forall t \in 1,\ldots,25 \tag{6}$$

$$Q(k,t) \ge Q(k,t-1) \quad \forall k \in 1,\ldots,14 \quad \forall t \in 2,\ldots,25 \tag{7}$$

$$Q_{Total}(t) = \mathrm{Q_{min}} + \sum_{k=1}^{n} Q(k,t) \quad \forall t \in 1,\ldots,25 \tag{8}$$

where $Q(k,t)$ is the amount of water sold to the k-th economic sector during the t-th year of the project. $\mathrm{Q_{min}}$ represents the projected drinking water shortage in Kerman Province over the project's planning horizon. $\mathrm{Demand}(k,t)$ denotes the water demand of the k-th economic sector in year t, calculated using Eqs. 11–15. Parameter $\mathrm{p}(k)$ refers to the maximum tariff that can be imposed on the water price, the payment of which is considered economically justified for economic sector k (Table 2) and parameter $\mathrm{p_{pot}}$ is the price of drinking water. $Q_{Total}(t)$ represents the total flow rate of the conveyance line during year t of project operation.

Using an EIOA-based linear programming model, we estimate the water demand of the economic sectors under an optimal economic development scenario, assuming that local water extraction remains limited to its current level due to water scarcity in the region. The EIOA model, a well-known extension of the input–output analysis (IOA) developed by Leontief[37], is among the widely applied techniques in economics[38]. Since the IOA does not account for the actual consumption rates of resources, e.g., water and electricity, due to government subsidies, the EIOA has been formulated to facilitate resource consumption analysis. The early formulation of the EIOA model[39] suffers from a data provisioning problem, so Bullard and Herendeen[40] proposed a hybrid-

unit technique in the new formulation of the EIOA as a remedy to the problem. An EIOA model employs linear algebraic relations and uses the related data in the input–output tables to calculate the direct and indirect resource consumption for producing a given amount of a commodity[41]. Miller and Blair[33] thoroughly explain the EIOA model's derivation procedure. Equation 9 shows the model's main formulation:

$$(\mathbf{I} - \mathbf{A})^{-1} \times \mathbf{f} = \mathbf{x} \tag{9}$$

where **I** is the $n \times n$ identity matrix, n is the number of economic sectors, **A** is the $n \times n$ technical coefficients matrix derived from IO tables, the elements of which (a_{ij}) represent the direct use of input i for producing one unit of output j, **f** is the $n \times 1$ vector of final demands, and **x** is the $n \times 1$ vector of the outputs of the economic activities.

To develop the EIOA model for Kerman's economy, data were obtained from the Iran Energy Ministry's *Energy Balance Sheet Report* (IEM, 2022), Kerman's *Regional Accounting Report*, and the national input–output (IO) tables published by the Statistical Center of Iran (SCI)[42]. The SCI provides national IO tables for 91 economic sectors every ten years, whereas Kerman's accounting reports cover 72 sectors annually. Both datasets follow the *International Standard Industrial Classification* (ISIC) framework. The difference in sectoral coverage arises because certain economic activities are absent in Kerman Province. To ensure consistency and computational efficiency, the economic activities in both datasets were aggregated into 14 broader sectors (see Table 1). To regionalize the national Input–Output (IO) table, the Cross-Hauling Adjusted Regionalization Method (CHARM)[43] was employed. The application of this method relied on two foundational assumptions: First, it assumes that Kerman Province's technical coefficients matrix corresponds directly to that of the national economy. This premise is considered robust given the aggregation level and the general similarity of production technologies across Iran's provinces[44]. Second, the method assumes that the degree of cross-hauling at the regional level is proportional to the sectoral heterogeneity observed at the national level. This assumption is supported by similarities between regional and national macroeconomic indicators; specifically, Kerman's proportionate share of value-added and total output across most economic sectors, with the exception of Agriculture and Oil & Gas. Following the construction of the Kerman's Input–Output table, *Energy Balance Sheet Report*[45] data were uesed to calculate monetary values of water and electricity consumption through the implicit price technique. Finally, they were converted into physical quantities to establish the EIOA model for Kerman's economy.

The EIOA model is employed to identify the optimal development scenario for Kerman's economy. For this purpose, an EIOA-based linear programming (LP) model is formulated with the objective function of maximizing regional GDP, leveraging the key feature of the IO table that the sum of final demands equals GDP. The model's constraints account for production technology limitations (matrix **A**), the maximum exploitable capacity of resources, and the production capacity of economic activities. Specifically, the output vector (**x**) is replaced by the production capacity vector ($\mathbf{x}^*$), as shown in Eq. 9, yielding inequality 10, which defines the constraint set of the optimization problem.

$$(\mathbf{I} - \mathbf{A})^{-1} \times \boldsymbol{f} \leq \mathbf{x}^* \tag{10}$$

In inequality 10, the production capacity ($\mathbf{x}^*$) of all economic activities, except the water sector, is assumed to increase proportionally with the population growth rate (PGR = 1.6% per year) over the 25-year project lifetime, with the first-year production capacity set equal to the observed output. For the water sector, 250 MCM is added to the current supply capacity, representing the maximum volume of water that can be transferred through the KDCL project. In addition to the constraint set defined in inequality 10, further constraints are required to ensure that residential water and electricity demands are met during the project's lifetime. Since these demands increase proportionally with population growth, Eqs. 13 and 14 are incorporated into the model to satisfy residential demand for water and electricity.

Maximizing the objective function of the model (Eq. 11) subject to the defined constraints yields a solution in which the water deficit is offset by reductions in the agricultural sector, as other sectors are more efficient water users. However, given the social tensions, reducing agricultural water consumption in the short term may not be practically possible. Evidence for this is the failure of Iranian Ministry of Energy's plan to reduce agricultural activity by up to 20% by 2040 to conserve groundwater resources[46]; after approximately a decade, agricultural activity has not decreased, and some aquifers have even experienced increased groundwater withdrawals by local farmers. Therefore, Eq. 15 is introduced as an additional constraint to prevent the model from reducing agricultural activity below current levels. Equations 11–15 together constitute the final LP model for determining Kerman's optimal economic development scenario. Considering limitations in production capacities and water resources, the model identifies the production levels of economic activities that maximize regional GDP, with the final demands of the economic activities ($f_k\,(t)$) serving as the model's decision variables.

Water (Water)	Electricity (Elc)	Clothes & textile (Cloth)	Wood and paper (Wood)	Agriculture (Agr)	Mining (Min)	Food-processing (Food)
Transport equipment (Trans)	Metal products (Metal)	Ceramic (Ceramic)	Chemical, oil and gas (Petrol)	Construction (Const)	Service (Srv)	Other industries (Other)

Table 1. Aggregated economic sectors based on national datasets.

$$MaxGDP(t) = q_{\mathrm{Water}}(t) \times \mathrm{p}_{\mathrm{Water}} + q_{\mathrm{Elc}}(t) \times \mathrm{p}_{\mathrm{Elc}} + \sum_{k=m+1}^{n} f_k(t) \tag{11}$$

Subject to:

$$(\mathbf{I} - \mathbf{A})^{-1} \times \boldsymbol{f}(t) \leq \mathbf{x}^{*}(t) \tag{12}$$

$$q_{\mathrm{Water}}(t) = (1 + \mathrm{PGR})^{\mathrm{t}} \times \mathrm{q}_{\mathrm{Water}}^{\mathrm{Current}} \tag{13}$$

$$q_{\mathrm{Elc}}(t) = (1 + \mathrm{PGR})^{\mathrm{t}} \times \mathrm{q}_{\mathrm{Elc}}^{\mathrm{Current}} \tag{14}$$

$$f_{Agr}(t) \geq \mathrm{f}_{\mathrm{Agr}}^{\mathrm{current}} \tag{15}$$

where q_{Water} and q_{Elc} are the final demands for water and electricity in physical units, and $\mathrm{p}_{\mathrm{Water}}$ and $\mathrm{p}_{\mathrm{Elc}}$ are the corresponding consumer prices (household tariffs). n denotes the total number of economic activities, and m represents the number of activities for which transactions are recorded in physical units (i.e., the water and energy sectors). f_k is the final demand for product k in monetary units (excluding the water and energy sectors). Additionally, $\boldsymbol{f}$ and $\mathbf{x}^{*}$ are the vectors of final demand and production capacity for the economic sectors, respectively. t indicates the projected year, and the superscript "Current" refers to the value of the corresponding parameter for the base year.

Solving the linear optimization problem (Eqs. 11–15) determines the optimal capacity expansion for all economic sectors during the project's operational years and allows estimation of the corresponding water demand ($\mathrm{Demand}\,(k,t)$) to reach this expansion based on sectors' water use intensity. As noted earlier, when water supply exceeds the demand of low water-intensive sectors, sectors with higher water use intensity but lower willingness to pay (WTP) become the potential customers for the transferred water. For instance, economic activities with low water intensity, such as the automobile industry, use water as a minor input and can tolerate higher water prices without significantly affecting production costs; however, their overall water demand remains small. Any remaining water exceeding the demand of these high-WTP sectors can then be allocated to lower-price sectors with higher water intensity, up to the point where the consumers' WTP be higher than the marginal cost of water supply through the KDCL project. This effectively allows water to be offered at different prices to different users, resembling a pay-as-bid market for water.

To estimate the WTP of different economic sectors based on their water use intensity derived from the EIO table, we reviewed water sale contracts—specifically, effluent from wastewater treatment plants—between several production units and INWW Eng. Co. This review indicated that the lower the contribution of water to an economic activity's total inputs, the greater the amount the activity is willing to pay for water (Fig. 3). Accordingly, we assume that the WTP for water in Kerman's economic sectors can be estimated from their water use intensity. Based on this assumption, the data obtained in this review were generalized using a curve-fitting approach to derive a functional relationship between the sectors' water use intensity and their WTP. Several functional forms were tested, and a logarithmic function with the highest R^2 value (0.937) was selected (Fig. 3). This function was then used to determine the water tariff for each economic sector, as reported in Table 2**.** It should be noted that, to ensure real-world applicability, the estimated WTP values were adjusted for each economic sector.

Cost estimation module

Fluid mechanics equations, technical standards, and official project price lists are applied in this module to develop a model for estimating the project's implementation costs (Eqs. 16–21) and operation costs (Eqs. 22–35). As a sub-module of the CBA optimization framework (Eqs. 1–4), this model determines the optimal values of the project's design variables, aiming to minimize total costs for a given volume of water transfer. While design variables such as pipe diameters and pump station characteristics are optimized by the model, other parameters—such as the conveyance route layout and pipe material—are assumed to be fixed and predetermined according to the initial plan proposed by the project's client (IWPC). In the preliminary design, steel pipes with a maximum working pressure of 40 bar were identified as the best techno-economically option for the KDCL project. Since steel pipes are manufactured in standard discrete diameters, pipe diameter cannot be treated as a continuous decision variable.

The layout of the project's conveyance line suggested through the feasibility studies conducted by IWPC (Fig. 1) was adopted in this study for the optimal hydraulic design of the KDCL system. Figure 4 shows the profile of the conveyance line. Since the highest and lowest elevations along the route represent the critical points, the conveyance path is divided into seven main segments corresponding to the major depressions and reliefs in the profile. Each segment is divided into four subsegments, allowing each to have a distinct pipe diameter—resulting in 28 segmental diameters defined as decision variables. The lengths of the subsegments are optimized, with their total length constrained to equal that of the corresponding segment. Pressure reducing valves (PRVs) and pumping stations can be installed only at the beginning of each subsegment. Due to the presence of binary variables, increasing the number of subsegments significantly raises computational costs while having negligible impact on the optimal solution as discussed in the results and discussion section. The optimization model determines the optimal values of all the design variables—including pipe diameters, flow velocities, and the placement and specifications of pump stations and PRVs—to maximize the objective function (net benefit).

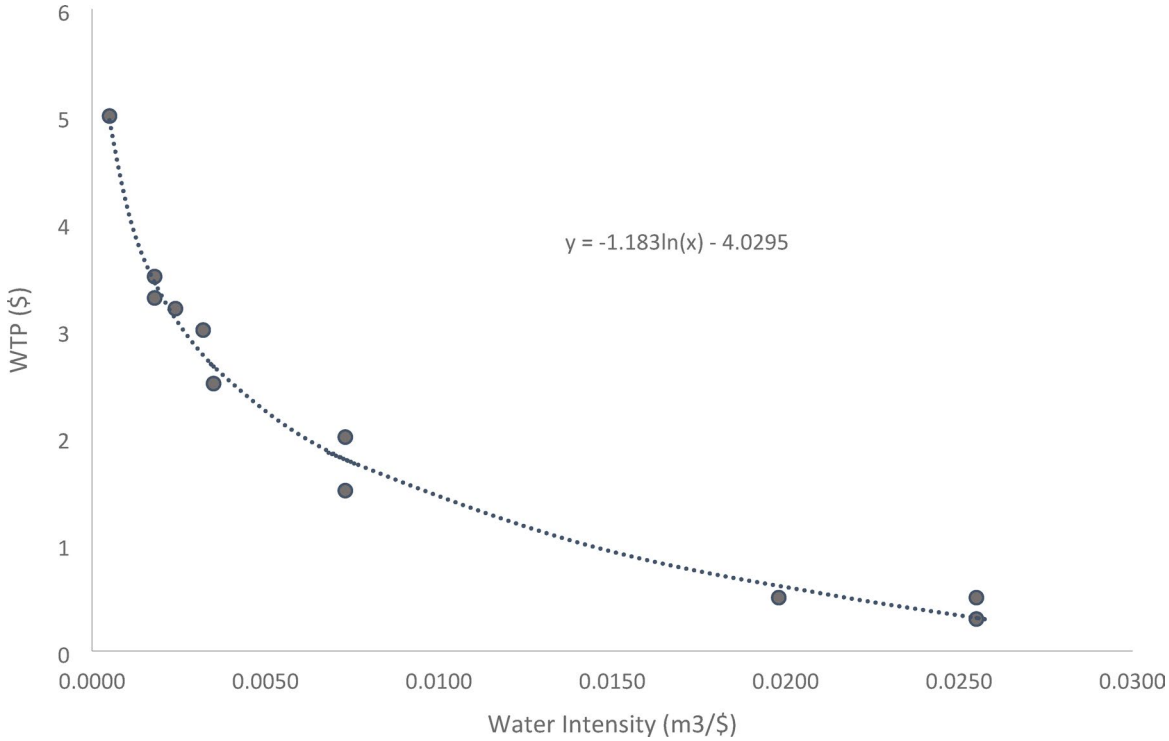


Fig. 3. Relation between the water intensity and the WTP for the water in the observed production units in the survey conducted by INWW Eng. Co.

Economic sector	Water intensity	Imposed tariff based on estimated WTP for One M^3 water
Water	0.0128 (m^3/m^3)	$0.5
Electricity	0.0009 (m^3/kWh)	$1.0
Cloth industry	0.0017 (m^3/$)	$4.0
Wood industry	0.0032 (m^3/$)	$2.5
Agriculture	3.0102 (m^3/$)	$0.1
Mining	0.0219 (m^3/$)	$1.0
Food industry	0.0029 (m^3/$)	$3.0
Automobile and industrial machine	0.0005 (m^3/$)	$6.0
Metal industry	0.0035 (m^3/$)	$2.0
Ceramic industry	0.0073 (m^3/$)	$1.5
Oil, gas and chemical industry	0.0018 (m^3/$)	$4.0
Construction	0.0198 (m^3/$)	$1.0
Service	0.0255 (m^3/$)	$0.5
Others	0.0024 (m^3/$)	$4.0

Table 2. Water intensity and imposed tariff for water based on the estimated WTP of the economic activities.

Equations 16–21 define the implementation cost estimation module of the optimization model. This module calculates the project's primary implementation costs, including the cost of piping, pump stations, and control valves. Additional implementation costs are incorporated as a percentage of the primary cost, based on benchmarks from similar projects. Table 3 , derived from the base price list of the Plan and Budget Organization of Iran, indicates the purchase price per kilometer of steel pipes across various diameters. Using the data from this table, Eq. 17 computes the total pipelaying costs. Equations 18 and 19 estimate the costs of pump stations and pressure-reducing valves (PRVs), respectively. These equations were derived by fitting curves to cost data from comparable cases: Eq. 18 relates pump station cost to power consumption, whereas Eq. 19 estimates the cost of purchasing and installing a PRV as a function of its diameter.

$$Cost^{Imp} = \mathrm{AF_{Imp}} \times \left(\sum_{\mathrm{i=1,j=1}}^{\mathrm{i=7,j=4}} Cost_{Pipe}\,(i,j) + \sum_{\mathrm{i=1,j=1}}^{\mathrm{i=7,j=4}} Cost_{Pump}^{Imp}\,(i,j) + Cost_{Valve}\,(i,j) \right) \quad (16)$$

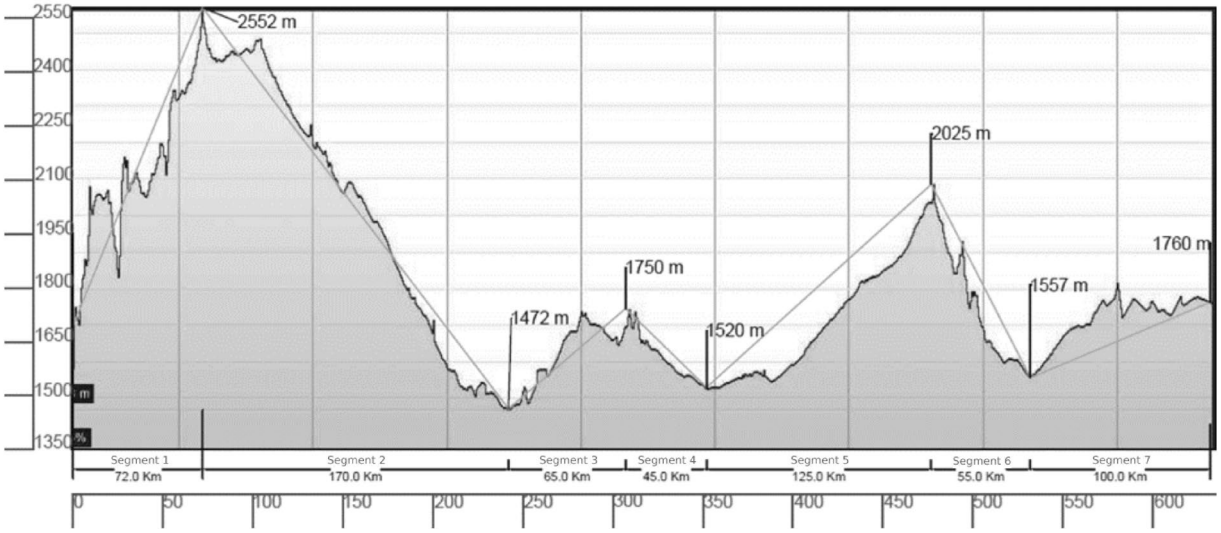


Fig. 4. The profile of the path suggested by IWPD Co.

Pipe diamter (mm)	1000	1200	1400	1600	1800	2000
Pipe cost ($/Km)	266,356	358,467	452,673	570,699	746,070	1,023,759

Table 3. Purchase price per kilometer of steel pipes across various diameters

$$Cost_{Pipe}(i,j) = \sum_{f=1}^{6} z_1(i,j,f) \times \mathrm{Cost}_{\mathrm{Pipe}}^{\mathrm{km}}(f) \times L(i,j) \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{17}$$

$$Cost_{Pump}^{Imp}(i,j) = 176823 \times Pump_{EP}(i,j)^{0.4} \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{18}$$

$$\mathrm{Cost}_{Valve}(i,j) = z_2(i,j) \times (4.4 \times D(i,j) - 26) \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{19}$$

$$D(i,j) = \sum_{f=1}^{6} (z_1(i,j,f) \times \mathrm{Dia}(f)) \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{20}$$

$$\sum_{f=1}^{6} z_1(i,j,f) = 1 \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{21}$$

In the above equations, i denotes the index of the main segments (seven in total), and j represents the index of the subsegments (four per segment). The variables $Cost_{Pipe}(i,j)$, $Cost_{Pump}^{Imp}(i,j)$, and $Cost_{\mathrm{Valve}}(i,j)$ refer to the costs of piping, pump stations, and PRVs in subsegment (i,j), respectively. Their sum constitutes the project's total implementation cost, denoted as $Cost^{Imp}$. The discrete index f corresponds to the pipe diameter $\mathrm{Dia}(f)$, ranging from 1200 to 2000 mm (f = 1,…,6). $\mathrm{Cost}_{\mathrm{Pipe}}^{\mathrm{km}}(f)$ represents the piping cost per unit length, including expenses for purchase, lining, coating, excavation, and backfilling for the f-th standard diameter. $\mathrm{Len}(i)$ and $L(i,j)$ denote the pipe lengths of segment (i) and subsegment (i,j), respectively. $\mathrm{Pump}_{\mathrm{EP}}(i,j)$ is the power of the pump installed at the beginning of subsegment (i,j). $D(i,j)$ is the diameter of the pipe used in subsegment (i,j), and $z_1(i,j,f)$ is the binary variable corresponding to the f-th standard diameter of the pipe used in subsegment (i,j), resulting in $7 \times 4 \times 6 = 168$ binary variables in total. Finally, $\mathrm{AF}_{\mathrm{Imp}}$ is a parameter accounting for additional implementation costs—such as equipment monitoring—expressed as a percentage of the project's primary implementation costs.

The primary decision variables in the cost estimation module are $D(i,j)$, which link operation costs to implementation costs. Since $D(i,j)$ takes discrete values, binary variables $z_1(i,j,f)$ are employed to ensure that the model selects only one value from the available standard diameters, as defined in Eqs. 20–21. Selecting pipes with larger diameters increases implementation costs but reduces operation costs due to lower frictional head losses and, consequently, lower electrical energy consumption. The electricity cost for pumping constitutes a major component of the operation costs. Considering the number of pumps, their power, and the efficiency of pumping stations, the operation cost estimation module calculates the project's total electrical energy consumption (Eqs. 22–35). It is noteworthy that to reduce computational cost of the model, the average flow

velocity over the operation period, $\overline{V}(i,j)$, is used in calculating the frictional head loss in Eqs. 26–27. The equations used to estimate the operation costs are summarized as follows:

$$Cost^{Opr}(t) = \mathrm{AF_{Opr}} \times \mathrm{Cost_{Elc}} \times 24 \times 365 \times Q_{total}(t) \times \sum_{i=1,j=1}^{i=7,j=4} Pump_{EP}(i,j) \quad \forall t \in 1,\ldots,25 \tag{22}$$

$$Pump_{EP}(i,j) = \frac{\rho \times \mathrm{g} \times Pump_H(i,j)}{1000 \times \mathrm{Pump_{Ef}}} \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{23}$$

$$H(i,j) = Pump_H(i,j) + (1 - z_2(i,j) \times \delta(i,j)) \times H(i,j-1) - H_s(i,j) - H_f(i,j) \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{24}$$

$$H_s(i,j) = \mathrm{Slope}(i) \times L(i,j) \tag{25}$$

$$H_f(i,j) = \frac{6.78 \times \bar{V}(i,j)^{1.85}}{\left(\frac{D(i,j)}{1000}\right)^{1.16} \times \mathrm{C}^{1.85}} \times L(i,j) \times 1000 \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{26}$$

$$\bar{V}(i,j) = \sum_{t=1}^{25} \frac{V(i,j,t)}{25} \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{27}$$

$$V(i,j,t) = \frac{Q_{total}(t)}{\frac{3.14}{4} \times \left(\frac{D(i,j)}{1000}\right)^2} \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \quad \forall t \in 1,\ldots,25 \tag{28}$$

$$H(1,0) = 0 \tag{29}$$

$$H(i,0) = H(i-1,4) \quad \forall i \in 2,\ldots,7 \tag{30}$$

$$\sum_{j=1}^{4} L(i,j) = Len(i) \quad \forall i \in 1,\ldots,7 \tag{31}$$

$$H(i,j) \le \mathrm{H^{max}} \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{32}$$

$$Pump_H(i,j) + (1 - z_2(i,j) \times \delta(i,j)) \times H(\mathrm{i,j-1}) \le \mathrm{H^{max}} \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{33}$$

$$Pump_H(i,j) \le \mathrm{H^{max}} \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \tag{34}$$

$$\mathrm{V_{min}} \le V(i,j,t) \le \mathrm{V_{max}} \quad \forall j \in 1,\ldots,4 \quad \forall i \in 1,\ldots,7 \quad \forall t \in 1,\ldots,25 \tag{35}$$

In the above equations, the parameter AF_{Opr} accounts for operation costs not precisely estimable at this stage, such as maintenance and repair. Based on similar projects, both AF_{Opr} and AF_{Imp} are set to 10%, with sensitivity analysis performed on these factors. $\mathrm{Cost_{Elc}}$ denotes the unit cost of electricity (kWh), and $Pump_H(i,j)$ represents the pumping head at the beginning of subsegment (i,j). The Parameters $\mathrm{Pump_{Ef}}$, ρ, g, and C denote pump efficiency, water density (997 kg/m^3), gravitational acceleration (9.8 m/s^2), and the Hazen–Williams roughness coefficient (130), respectively. $H(i,j)$ is the water head at the end of subsegment (i,j), while $H_s(i,j)$ and $H_f(i,j)$ correspond to static and frictional head losses, respectively. $\mathrm{Slope(i)}$ and $\mathrm{H^{max}}$ denote the average ground slope in segment (i) and the maximum working pipe pressure. $V(i,j,t)$ is the flow velocity in subsegment (i,j) in year t, constrained within $\mathrm{V_{min}}$ (0.6 m/s) and $\mathrm{V_{max}}$ (2.5 m/s), the design limits for flow velocity. The average velocity over the project's operational period, $\overline{V}(i,j)$, is used to reduce computational effort in solving the optimization problem. Binary variables $z_2(i,j)$ determine whether a pressure reducing valve (PRV) should be installed at the beginning of subsegment (i,j), and the percentage of pressure reduction at a PRV is represented by the parameter δ, which ranges from 0 to 1.

In total, the optimization framework (Eqs. 1 to 4) handles 662 main decision variables. This includes 196 binary variables dedicated to pipe diameter selection (168) and PRV installation locations (28), as well as 84 continuous variables governing hydraulic parameters (such as pipe lengths, pressure reduction levels, and pumping power), and finally 350 continuous variables to identify the water allocation pattern across 14 economic sectors over the 25-year project lifetime. It should be noted that the model also incorporates numerous auxiliary variables, such as flow velocity in the pipes and hydraulic head at the end of sub-segments, whose values are strictly dependent on these primary decision variables.

Key fixed parameters within the model, including the Hazen–Williams coefficient, velocity bounds, and working pressure limits were established based on Iran's standard for designing water supply systems, known as Publication 3-117[47] and the steel pipes specifications provided in the main manufacturer's catalog[48]. The above MINLP model was solved using the ANTIGONE solver, a robust global optimization tool available in GAMS. ANTIGONE employs the Branch-and-Reduce-and-Cut (BARC) algorithm, which integrates branch-and-bound, reduced gradient, and cutting-plane methods to efficiently search for the global optimum of non-convex and nonlinear problems. This approach iteratively partitions the problem into subproblems and refines the solution space until the global minimum is identified. The solver is particularly effective for highly nonlinear and non-convex formulations where conventional techniques often fail to converge to optimal solutions. To

guarantee a rigorous and globally optimal solution, the solver's relative optimality tolerance (optcr) was set to 0.1 and the absolute optimality tolerance (optca) was set to 0.0.

Results and discussion

The results of of this study reveal several key quantitative outcomes. Under the optimal solution, 117.7 MCM/year of water is transferred through the KDCL system, of which 111.5 MCM/year (≈94%) is allocated to drinking water supply and 6.2 MCM/year to economic sectors. The estimated marginal cost of transferred water is approximately \$0.7 per m^3, which acts as a threshold determining whether supplying water to specific sectors is economically justifiable from the investor's perspective. Consequently, water supply to agriculture, service, and water provision sectors is excluded in the optimal allocation because their willingness to pay does not cover this cost. Despite allocating only a small portion of the system capacity to economic activities, the project is able to satisfy more than 85% of the incremental water demand of most industrial sectors at reasonable cost. The following subsections present the detailed outcomes of the two optimization stages and discuss their economic and hydraulic implications.

The proposed framework involves two levels of optimization: first, the outcome of a simple EIOA-based LP model used to determine the production capacity expansion in the economic sectors that would maximize regional GDP under conditions where further extraction from local water resources is not possible and the full capacity (250 MCM/year) of the water transfer project is utilized. Accordingly, the water demands of the economic sectors required to support their projected production capacity expansion are calculated. Second, the outcome of a more complex MINLP model solved to identify supplying what portion of the water demand determined in the first step is economically justifiable from the project investor's perspective, as well as optimal hydraulic design of the KDCL project.

Table 4, derived from solving the EIOA-based LP model (Eqs. 11–15), presents the calculated water demand of economic sectors required to achieve the optimal production capacity expansion—approximately 111.5 MCM/year out of the total 250 MCM/year should be reserved to meet drinking water demand. Table 4 indicates that the capacity expansion of all economic sectors almost follows the expected trajectory in line with the PGR, except for the agriculture sector, where limited water resources constrain expansion. In fact, the production capacities of less water-intensive industries are fully utilized to achieve the highest possible GDP growth under water-scarce conditions. It is also notable that as approaching the final years, the economic growth rate gradually declines due to increasing water shortages, decreasing from approximately 1.6% in the first year to about 1.2% in the final year.

Using the estimated water demands of the economic activities (Table 4) and their willingness to pay for water (Table 2), Eqs. 1–4 were solved in conjunction with the benefit and cost estimation modules Eqs. 5–8 and 16–35 to determine the optimal water allocation among economic sectors and the optimal hydraulic design of the project. In the optimal solution, out of the total feasible water transfer capacity of 250 MCM/year, 117.7 MCM/year is utilized—111.5 MCM/year allocated to drinking water supply and 6.2 MCM/year to economic sectors. Although only 5.3% of the conveyance system's total transfer capacity is assigned to economic activities, the project can supply over 85% of the increase in water demand among these sectors at a reasonable cost—excluding the service, water provision, and agriculture sectors. Table 5 presents the percentages of the water demands listed in Table 4 that can be met by the KDCL project at the obtained optimal solution.

Year													
Activity	1	3	5	7	9	11	13	15	17	19	21	23	25
Water	1.639	2.012	2.012	2.012	2.012	2.012	2.012	2.012	2.012	2.012	2.012	2.012	2.012
Elec	0.073	0.179	0.267	0.357	0.450	0.546	0.645	0.747	0.852	0.961	1.073	1.190	1.309
Cloth	0.001	0.002	0.003	0.004	0.005	0.007	0.008	0.009	0.011	0.012	0.014	0.015	0.017
Wood	0.002	0.006	0.010	0.015	0.019	0.024	0.029	0.034	0.039	0.044	0.050	0.055	0.061
agriculture	71.596	122.076	121.957	115.856	109.559	103.058	96.348	89.421	82.271	74.890	67.271	59.407	51.289
Mining	0.167	0.508	0.860	1.223	1.599	1.986	2.385	2.798	3.224	3.664	4.118	4.586	5.070
Food	0.007	0.020	0.034	0.049	0.064	0.079	0.095	0.111	0.128	0.146	0.164	0.182	0.202
Auto and …	0.001	0.002	0.003	0.005	0.006	0.008	0.009	0.011	0.013	0.014	0.016	0.018	0.020
Metal	0.028	0.086	0.146	0.208	0.272	0.338	0.406	0.476	0.549	0.623	0.701	0.781	0.863
Ceramic	0.009	0.029	0.050	0.070	0.092	0.114	0.137	0.161	0.186	0.211	0.237	0.264	0.292
Oil, Gas, …	0.015	0.046	0.078	0.110	0.144	0.179	0.215	0.253	0.291	0.331	0.372	0.414	0.458
Construction	0.033	0.100	0.169	0.240	0.314	0.389	0.468	0.549	0.633	0.719	0.808	0.900	0.995
Service	2.491	7.594	12.861	18.298	23.910	29.703	35.683	41.856	48.229	54.806	61.596	68.605	75.841
Others	0.001	0.003	0.004	0.006	0.008	0.010	0.012	0.014	0.016	0.019	0.021	0.023	0.026
Sum	76.062	132.662	138.453	138.453	138.453	138.453	138.453	138.453	138.453	138.453	138.453	138.453	138.453

Table 4. Water Demand (MCM/year) of the economic sectors over the KDCL project's life time to reach the optimal prodution capacity expantion undre water scarce condintion (extraction from local water resources is not possible and the full capacity (250 MCM/year) of the water transfer project is utilized) __numbers privided biyearly due to lack of space).

Year													
Activity	1	3	5	7	9	11	13	15	17	19	21	23	25
Water	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Elec	100%	100%	100%	100%	100%	100%	100%	100%	90%	80%	72%	65%	59%
Cloth	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Wood	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Agriculture	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Mining	100%	100%	100%	100%	100%	100%	100%	100%	92%	81%	72%	65%	59%
Food	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Auto and …	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Metal	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Ceramic	100%	100%	100%	100%	100%	100%	100%	100%	87%	76%	68%	61%	55%
Oil, Gas, …	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Construction	100%	100%	100%	100%	100%	100%	100%	100%	88%	78%	69%	62%	56%
Service	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Others	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Table 5. The percentages of water demands of the economic sectors that can be met by the KDCL project (numbres are given biyearly).

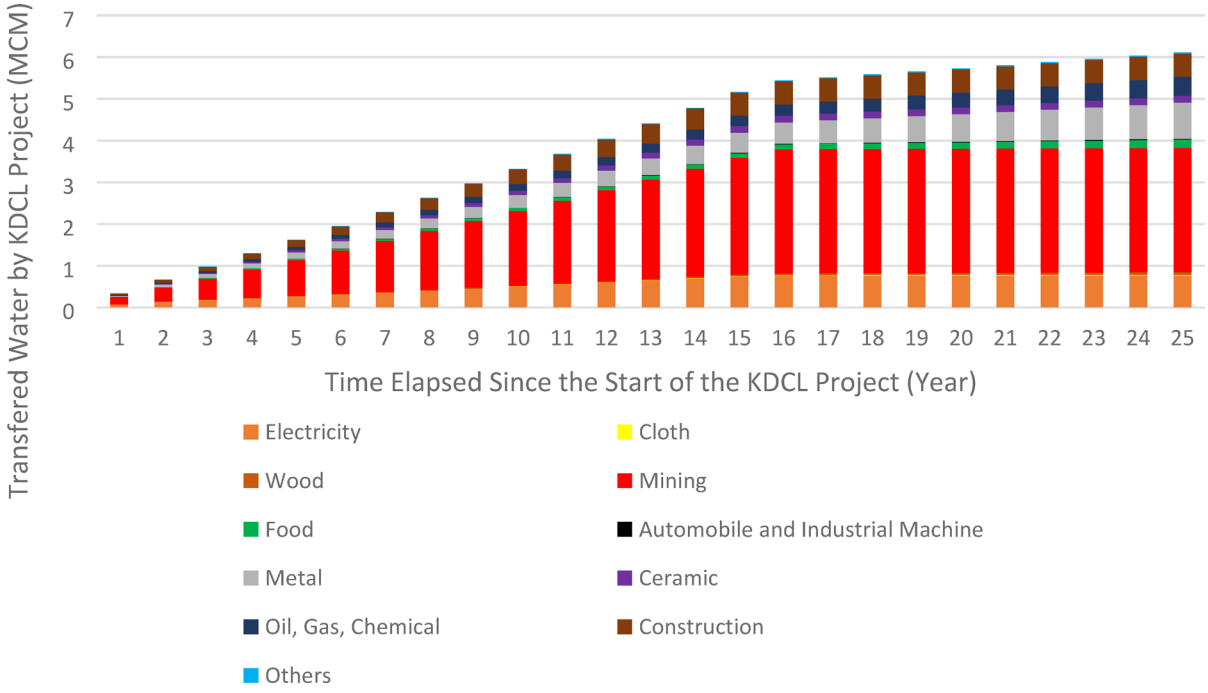


Fig. 5. Optimal temporal pattern of water allocation of the KDCL project.

The results indicate that the marginal cost of water production through the optimally designed KDCL conveyance line is estimated at $0.7 per cubic meter on avarage during the projet's life time. This represents the minimum price that must pay to make water supply economically justifiable from the project investor's perspective. Consequently, supplying water to economic activities that cannot afford the marginal cost of water —specifically, the agriculture, water provision, and service sectors—is not economically justified. Moreover, the full allocation of water to the electricity, mining, construction, and ceramic sectors is feasible only during the early years of the project, after which declining water availability constrain their economic growth. Figure 5 illustrates the allocation pattern proposed by the model to maximize the net benefit of the KDCL project. Despite having sufficient capacity for additional water transfer in the early years, the demands of certain sectors are not fully met during this period. This outcome may be due to Eq. (7), which restricts allocating less water to a sector compared to the previous year. Adopting this policy prevents capacity expansion without considering the sector's long-term water requirements. Another reason behind this outcome is that the water prices some sectors are willing to pay do not even cover operational costs.

Since the primary objective of the project is to address the drinking water deficit in Kerman Province, approximately 94% of the total transmission capacity (111.5 MCM) is allocated to meeting the sector's final water demand. Accordingly, the project's net benefits for private investors largely depend on the tariff charged to households for the transferred water. Considering a price of $\$1/m^3$, set slightly above the marginal cost of water supply to provide incentives for private sector investment, the project's net profit was evaluated under two different scenarios: (1) supplying water solely for domestic use, and (2) supplying water to both domestic and economic sectors. In the first scenario, the present value (PV) of total project costs is estimated at approximately \$806.1 million, while in the second scenario, it slightly increases to \$823.6 million. Nevertheless, due to the additional revenue generated from water sales to economic sectors, the PV of the project's net benefit rising from \$384.5 million to \$408.7 million in the second scenario. The PV of payments made by economic activities for water is estimated at \$41.6 million, increasing the project's net benefit by approximately 6.3%. Notably, this contribution is calculated using an 8% discount rate, in line with standard practice in Iran. A lower discount rate would further increase the contribution of economic activities, as most payments are scheduled for the later years of the project.

Water price is a critical determinant for water-intensive economic activities, as it directly affects their profitability, and in some sectors, limited access to water at economically viable prices can significantly constrain or even halt further growth. Accordingly, ensuring an adequate and reliable water supply for certain sectors—such as services and electricity—is essential, even if the payments received do not fully cover production costs, as these sectors are fundamental to public welfare. Moreover, given the prevailing socioeconomic conditions in Kerman, households cannot be charged high tariffs for drinking water. Despite the strong rationale for providing subsidized water to these consumer groups, this study assumes that water supplied through the KDCL project is allocated only to consumers who are able to cover its cost, subject to the constraint that drinking water demand must be fully met. However, policymakers may ultimately choose to subsidize water for specific groups, such as domestic users. Another point that should be taken into consideration is that, in this study— The prevailing conditions in the non-conventional water market in Iran—water pricing for economic users is assumed to occur through contractual arrangements rather than through a standard market-clearing mechanism. This system resembles a pay-as-bid structure in which water prices vary across users. Consequently, some sectors pay below the equivalent marginal price of water, while others pay above it. This pricing framework enables water to be allocated to a larger number of economic activities, thereby contributing to higher regional GDP.

Figure 6 presents the optimal hydraulic design of the KDCL, including pipe diameters, the locations of pump stations and PRVs, and the energy line. As shown, the conveyance system is divided into seven main segments with a total length of 650 km. Although the pipe diameter could range between 1200 and 2400 mm, the optimized design employs diameters varying from 1600 to 1800 mm, selected to balance implementation costs and head losses along the path. Four main pumping stations are installed to overcome the elevation differences along the conveyance line, with pumping heads ranging from 152 to 400 m. A pressure reducing valve (PRV) with a head loss of about 200 m is placed downstream of the highest elevation point (2552 m) to prevent excessive pressure in subsequent subsegments. Additionally, two others passive PRVs are installed to dissipate excess static pressure in the event of flow interruption. The energy line indicates that the hydraulic gradient remains within a safe range throughout the system, ensuring efficient energy use and stable operation. This configuration minimizes both energy consumption and implementation costs while maintaining adequate pressure across all segments of the conveyance system. The results also confirm the appropriateness of dividing each main segment into four

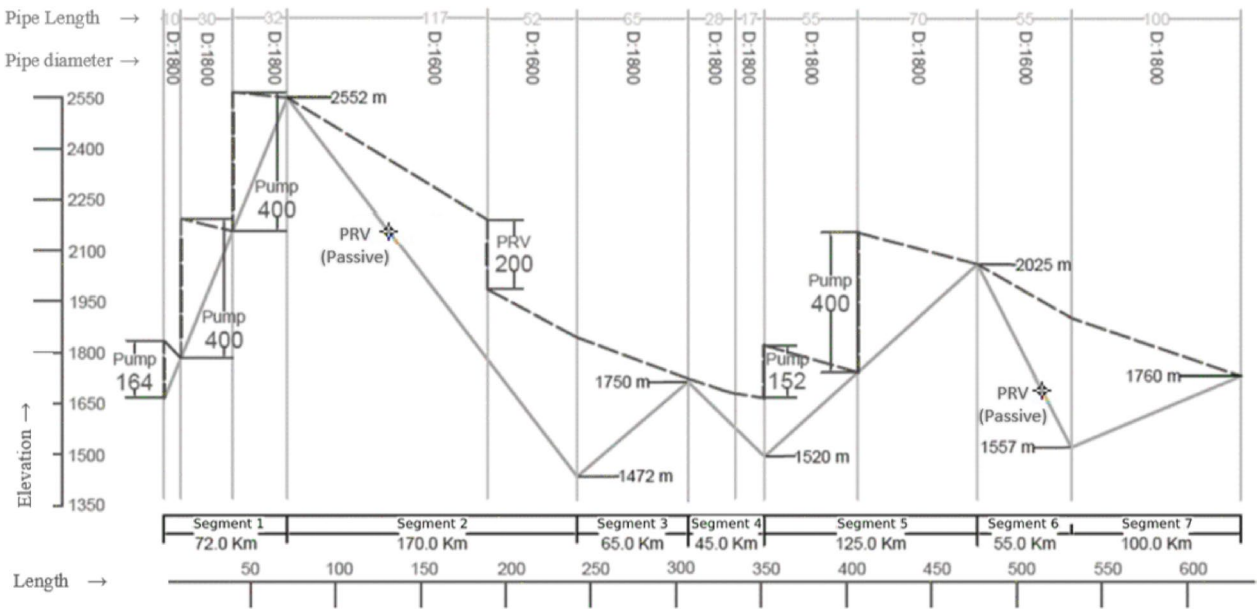


Fig. 6. Optimal hydraulic design of the KDCL project and the associated energy line resulted.

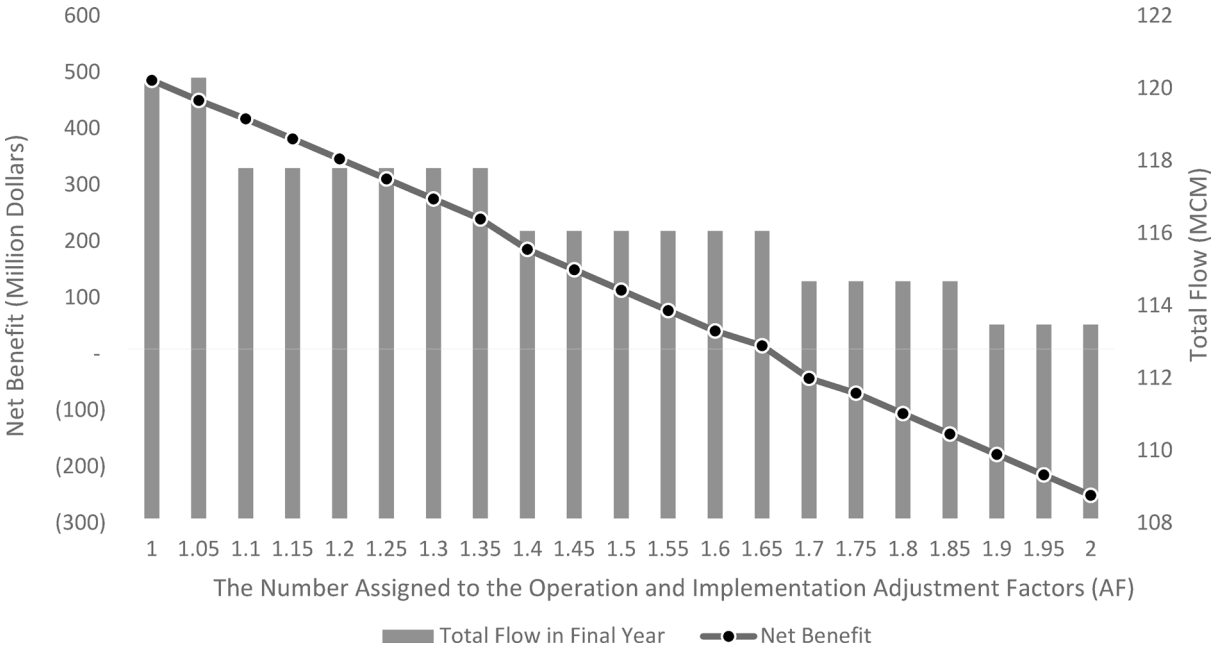


Fig. 7. Sensitivity analysis with respect to $\mathrm{AF}_{\mathrm{Opr}}$ and $\mathrm{AF}_{\mathrm{Imp}}$ paramters.

subsegments, as discussed in "Modeling approach" section, since the optimal number of diameter transitions and installations of pump stations or PRVs per segment does not exceed three.

The results of the sensitivity analysis on the parameters $\mathrm{AF}_{\mathrm{Opr}}$ and $\mathrm{AF}_{\mathrm{Imp}}$ are presented in Fig. 7, which illustrates how changes in these two parameters affect the project's net benefit and the volume of water transferred. Increasing these adjustment factors corresponds to higher project costs; consequently, the project's net benefit decreases as their values rise. A similar pattern is observed for the volume of water transferred. However, while the project's net benefit declines continuously and gradually, the reduction in the transferred water volume is not gradual, showing several sudden drops. This indicates that, due to the imposed tariff structure, certain economic sectors may abruptly be excluded from the water supply plan once project costs exceed a specific threshold, making water supply to those sectors no longer economically justifiable for the investor of the project. Provided that the project's side costs exceed 65 percent of the main cost ($\mathrm{AF}_{\mathrm{Opr}}$ and $\mathrm{AF}_{\mathrm{Imp}}$ coefficients are more than 1.65), the project will no longer be profitable for the investor. In this case, either the price of drinking water or the tariffs sat for selling water to economic sectors (Table 2) must be increased to create an incentive for investment in this project.

Conclusion

High-cost inter-basin WCPs, despite their potential socio-environmental impacts, are sometimes the only reliable solution to severe water scarcity challenges. Ensuring a sustained water supply to Kerman Province—an arid region where water shortages significantly hinder development—has been a major concern for policymakers, leading to the initiation of the KDCL project to meet the province's drinking water demand in the first place. However, the high implementation and operational costs necessitate the involvement of economic sectors to help mitigate the financial burden while simultaneously addressing their own water shortage problem. This study presents a novel optimization framework for the optimal design of the long-distance KDCL project, simultaneously addressing the interrelated challenges of determining the project's optimal hydraulic configuration, water transfer volume, sectoral production capacity expansion, and water allocation among economic sectors. In the proposed approach, sectoral production capacities—and consequently, their water demands—are first determined using an EIOA-based linear programming model. The resulting values are then incorporated into a mixed-integer nonlinear programming (MINLP) model, solved using the BARC algorithm to achieve globally optimal design and allocation pattern in this project.

The results indicate that the participation of less water-intensive economic sectors in this large-scale drinking water supply project not only satisfies their water demands but also helps offset part of the project's costs. However, supplying water to highly water-intensive sectors—such as agriculture—through such a relatively high-cost project is not economically justified. The proposed method effectively determines the optimal water allocation among economic sectors, as well as their respective contribution shares to the project's benefits and costs. One of the key advantages of this approach lies in its ability to prescribe an optimal economic development strategy for regions facing water scarcity. Future research may focus on extending the economic component of the model into a more advanced models, such as a Computable General Equilibrium (CGE) model, or enhancing the EIOA model to account for dynamic economic conditions.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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